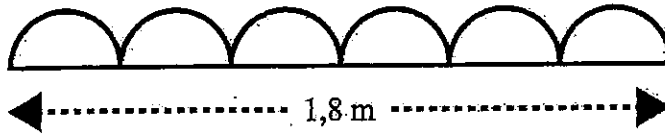
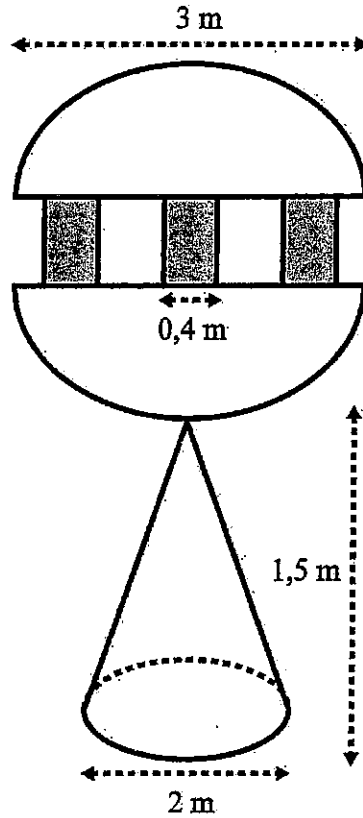


1. You have been hired to help design speed bumps for a busy road. The road is 1,8 m wide. Speed bumps are perfectly hemispherical and should be placed side by side so that 6 speed bumps fit exactly across the road as shown in the diagram below.



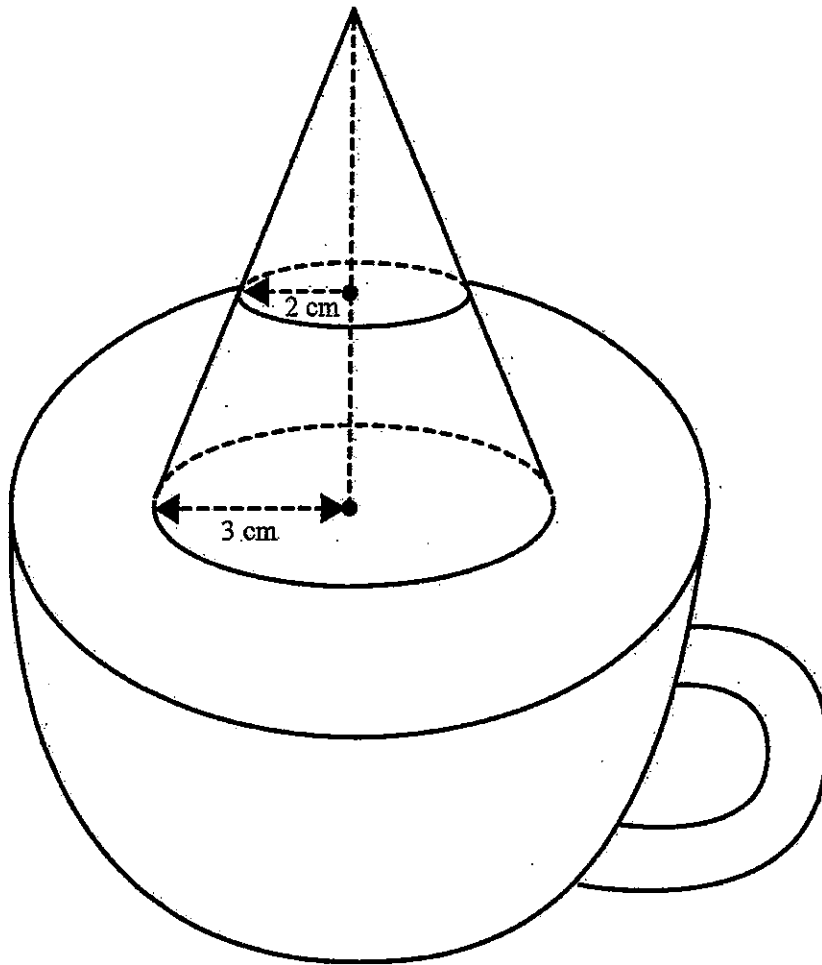
- 1.1 Determine how much metal (in cm^3) is required to make the speed bumps for a single point. (5)
- 1.2 You decide that every second bump should be painted with glow in the dark paint so that drivers can see them clearly at night. What is the total area of speed bumps that you have to paint? (4)

2. A friend of yours has opened a hamburger restaurant called Betty's Burger Bonanza. She has asked you for some help in designing a model hamburger on a cone to be displayed outside the restaurant. The model is to be made out of two foam hemispheres that each have a diameter of 3 m, three foam cubes 0,4 m wide and high, and a conical metal base with a height of 1,5 m and a diameter of 2 m.



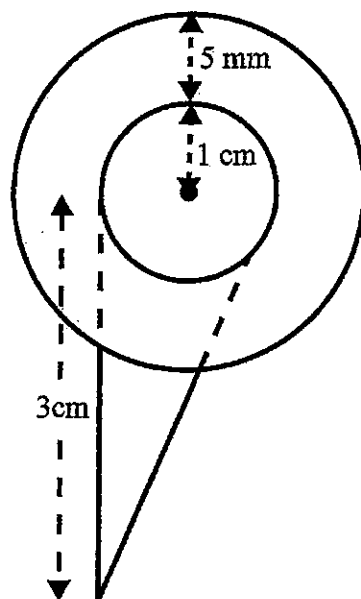
- 2.1 Your friend wants to paint the cone. What area does she have to paint? (5)
- 2.2 What is the volume of the burger and filling (i.e. how much foam is needed)? (5)

3. A coffee shop serves its jumbo cappuccino with a perfect 4 cm high cone of cream on top. The owner of the coffee shop wants to save money by using less cream. He decides to make the cone 2 cm shorter. Originally, the cone had a radius of 3 cm, and the shortened version will have a radius of 2 cm.



- 3.1 How much cream will be saved per cappuccino? (5)
- 3.2 The coffee cup holds 380 cm^3 of coffee and is perfectly hemispherical inside. Find the radius of the coffee cup. (3)
- 3.3 How much cream, correct to the nearest cm^3 , would be needed to completely cover the coffee with a 4 cm high cone of cream (i.e. the base of the cone is exactly the same as the surface area of the top of the coffee cup)? (2)

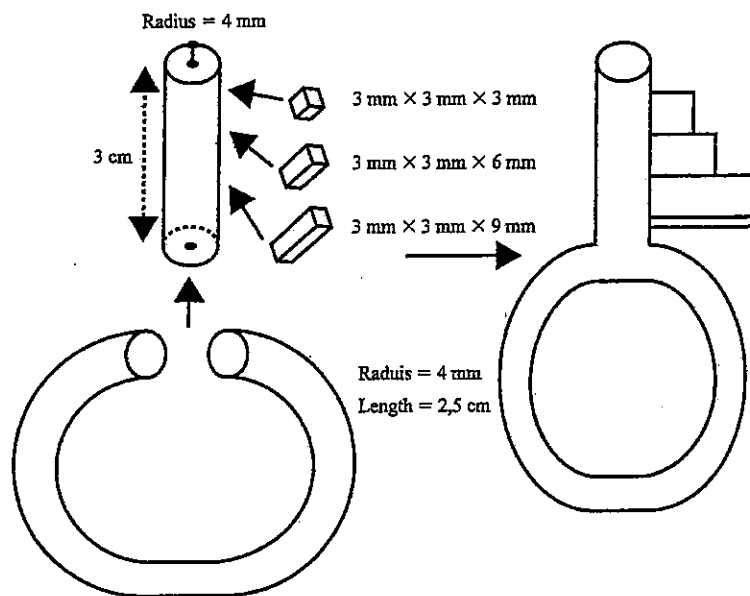
4. A friend of yours has started a business manufacturing hardware supplies. He has asked for some help in working out how to make nails. You work together and design a simple nail as shown below (viewed from above).



- Each nail is 3 cm long and has a diameter of 1 cm.
- Each nail has a rim that sticks out 5 mm and is 2 mm thick.
- Aside from the rim, each nail is a perfect right cone.
- All answers should be given in cm^3 .

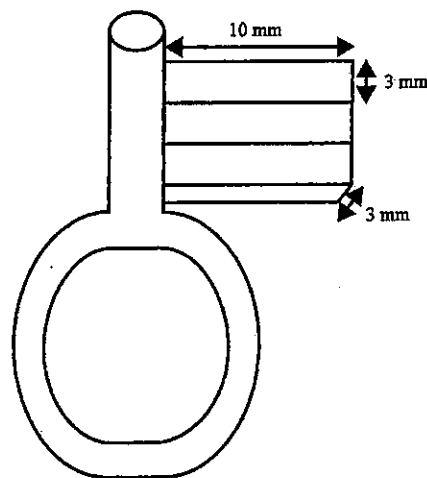
- 4.1 How much metal is required to make just the rim of the nail? (4)
- 4.2 Determine how much metal is required to produce each nail. (4)
- 4.3 If each nail is made 1 cm longer but is otherwise identical to the original design, how much extra metal is required to produce each nail? (3)

5. A locksmith is designing a simple key. The key is made as shown below. The loop at the bottom and the main stem of the key are cylinders and the teeth of the key are boxes.



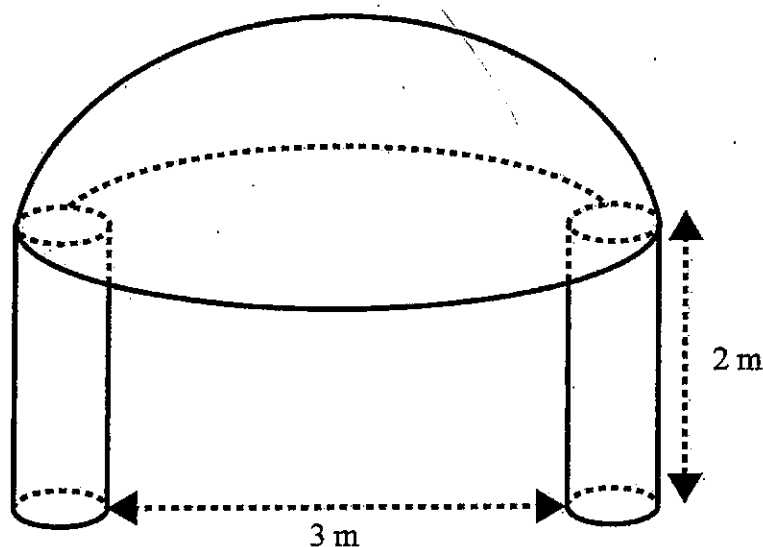
- 5.1 What volume (in mm^3) of metal is required for each key? (6)

Instead of sticking tiny pieces of metal together, the locksmith starts with the shape shown below and files it into the shape he requires. He melts down the filings and uses them to make other keys.



- 5.2 How much metal will be filed off when he makes one key? (3)
- 5.3 After how many keys will he have enough metal from the filings to make another key? (2)

6. You have been hired to help design a bus shelter. The shelter is made from two cylindrical pillars and a hemispherical shell as shown below.



- The top shell is 5 cm thick and hollow on the inside.
 - There must be 3 m of space between the pillars.
 - The pillars must each be 2 m tall and have a radius of 10 cm.
- 6.1 How much metal (in cm^3) is required to make the top shell? (5)
- 6.2 How much metal (in cm^3) is required to make the two pillars? (3)
- 6.3 You have to paint the bus shelter. Assume that only the outside of the hemispherical shell is painted and that the pillars are only painted around the sides (i.e. not on top or on the bottom). What is the area of the bus shelter that you will have to paint? (4)